Aayaann Kausar

AI/ML Researcher | LLM Post-Training & Multi-Agent Systems
Kolkata, West Bengal, India | +91 8583929615 | aayaannkausar@gmail.com | LinkedIn | GitHub

Summary

IIT Kharagpur undergraduate active in AI/ML research, LLM fine-tuning, and real-time speech AI. Skilled in Python, PyTorch, Hugging Face Transformers, QLoRA, PEFT, TRL, and LangGraph. Experienced in LLM evaluation, domain adaptation, and agentic AI research; technical stack includes FastAPI, Docker, PostgreSQL, Linux, Git, WebSockets, and deployment of real-world AI tools.

Education

Indian Institute of Technology (IIT) Kharagpur Kharagpur, India
Bachelor of Technology | ***Minor in Artificial Intelligence*** *2024 – Present*

Scholastic Achievements

JEE Main 2024 — Scored in the top **0.29%** of 1.4M+ candidates, among India's highest-scoring engineering aspirants nationwide.
JEE Advanced 2024 — Cleared the top **3.2%** of 250K+ aspirants, entering India's premier engineering entrance's elite tier.
WBJEE 2024 — Secured top **0.20%** among 113K+ candidates, one of West Bengal's strongest engineering exam scores overall.

Projects

Nova | Autonomous AI Co-Scientist | GitHub Jun 2026 – Jul 2026
*AI research platform extending **Google DeepMind's** Co-Scientist via LangGraph, literature grounding, hypothesis evolution, and HITL.*
- **Recreated Google DeepMind's** AI Co-Scientist in LangGraph, using literature, generation, reflection, ranking, and evolution.
- **Assessed** liver-fibrosis discovery, reaching **71.9/100** SHQI and **74.63/100** on the AI Co-Scientist Benchmark Score in final tests.
- **Measured 105** LLM calls and 210,620 tokens in 11.5 minutes, with HITL approval, LangSmith traces, and hypothesis diversity.

Real-Time Voice Translator | Multilingual Speech AI | Demo | GitHub Sep 2026 – Oct 2026
Deployed 7-language voice translator on Next.js, FastAPI WebSockets, Groq Whisper/GPT-OSS 20B, and ElevenLabs Flash v2.5 TTS.
- **Integrated** a **7-language** two-way **Whisper/GPT-OSS** translation pipeline via FastAPI WebSockets and ElevenLabs Flash TTS.
- **Implemented 20ms RMS VAD** with 600ms pause detection, polyphase resampling, turn-ID safeguards, and stale-output filtering.
- **Shipped** a **Next.js 16 AudioWorklet** UI on Vercel/Render with bilingual captions, live p50/p95 metrics, and 27 backend tests.

SFP | LLM Forgetting Mitigation | GitHub Feb 2026 – Oct 2026
Empirical study of PCA-ranked activation drift in Qwen2.5-1.5B to quantify catastrophic LLM forgetting and test SFP regularization.
- **Isolated** 32 ablation-ranked PCA axes in **Qwen2.5-1.5B**; top-ranked drift tracked forgetting ($R^2 = 0.933$) vs **0.850** random.
- **Formulated** rank-32 **SFP** feature loss, attaining 0.754 retention and 0.650 plasticity; 0.699 vs. 0.631 harmonic score (LwF).
- **Verified** a conditional forgetting bound in **Lean 4**; found **scale-sensitive** drift predictors failing at 135M but valid at 1.5B.

Qwen2.5 | QLoRA Fine-Tuning Study | GitHub Aug 2026 – Sep 2026
Empirical study of Qwen2.5-1.5B fine-tuning using 4-bit QLoRA, paired benchmark evaluation, complexity scaling, and failure analysis.
- **Fine-tuned Qwen2.5-1.5B** with **4-bit NF4 QLoRA** for 1,286 ML systems tasks using rank-16 adapters, TRL, PEFT, and 8-bit AdamW.
- **Quantified 30.42% ROUGE-L** gains over frozen Qwen on **161 held-out examples**; BERTScore F1 increased from 0.8187 to 0.8497.
- **Mapped** 48 technical topics and difficulty tiers; Staff ROUGE-L rose by **34.84%**, while 14.2 percent of tests regressed.

NexusAgent | Enterprise Decision Intelligence Platform | GitHub Aug 2026 – Sep 2026
Enterprise decision platform combining LangGraph agents, Hybrid RAG, guarded MCP NL2SQL, evidence review, plus scoped memory.
- **Elevated** hybrid RAG stack via Milvus, BM25/RRF and BGE, raising Hit@1 from **85.62%** to **96.25%** and MRR@5 to **98.12%**.
- **Secured** MCP NL2SQL via SQLGlot guards, read-only scopes, LIMITs/timeouts, evidence review, passing **28/28** security cases.
- **Orchestrated 3 LangGraph agent routes** with QA evidence review, validated by **1,941** unit and **323** offline integration tests.

Competition/Conference

HFT Market State Forecasting Challenge | Wunder Fund Sep 2025 – Nov 2025
Multivariate streaming sequence modeling forecasting high-frequency market state vectors across long 1,000-step sequences.
- **Ranked 9th globally** among 5000+ competitors worldwide, achieving **0.3920 R^2** final score and 0.3880 public leaderboard score.
- **Pretrained** a 4-layer, 8-head Transformer with Masked Autoencoders, temporal positional encodings, and median-IQR scaling.
- **Accelerated** streaming inference on single-core CPU via `torch.compile`, Numba kernels, and preallocated state memory buffers.

Skills & Expertise

Languages & Core: Python, C++, SQL, Bash, Linux, Git, NumPy, JavaScript, TypeScript, AudioWorklet, WebSockets

AI/ML & Research: PyTorch, Hugging Face, Transformers, PEFT, QLoRA, TRL, Lean 4, ROUGE-L, BERTScore F1, Whisper

Agents & Systems: LangGraph, LangChain, LangSmith, MCP, Hybrid RAG, FastAPI, Docker, Postgres, Next.js 16, ElevenLabs

Positions of Responsibility

Associate Member – Kharagpur Data Analytics Group (KDAG), IIT Kharagpur *Mar 2025 – Aug 2025*
- **Led** a **9-member team** to **1st place** in the KDAG Intra Hackathon, aligning members around multimodal medical AI pipelines.
- **Spearheaded** KDAC's **Elyx Life Hackathon**, leading outreach and execution for a **3-day AI-for-healthcare challenge**.

Coursework

Mathematics: Advanced Calculus | Probability and Statistics | Linear Algebra | Numerical Analysis
CS: Data Structures and Algorithms **AI:** Deep Learning Foundations | Reinforcement Learning | Foundations of LLMs